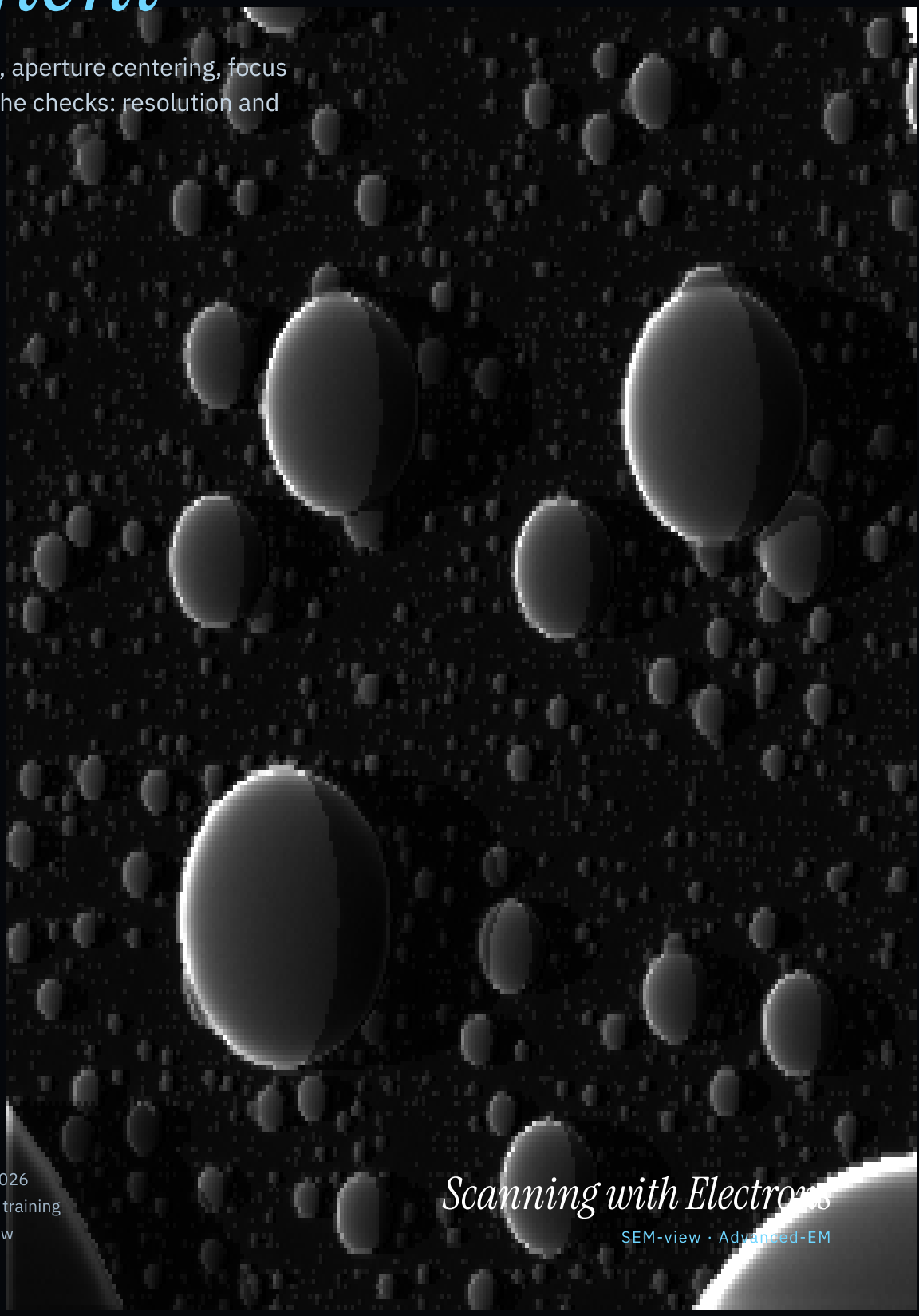
SEM column *alignment*

Gun, beam conditions, aperture centering, focus and stigmation, then the checks: resolution and magnification.



SOP-01, version 1.0, October 2026
For new users and for refresher training
Figures computed with SEM-view

Scope and preparation

This procedure aligns a field-emission SEM for high-resolution secondary-electron imaging. Do the steps in order: each assumes the previous ones are correct. Redo steps 3 and 4 whenever you change the accelerating voltage, the aperture or the working distance.

	Step	Adjustments	Checked with
1	Gun	None for a Schottky gun; flash for a cold FEG; filament saturation for tungsten or LaB_6	Emission current, extractor voltage, gun vacuum
2	Beam conditions	Voltage, probe current, aperture, working distance	The task (table on page 3)
3	Aperture centering	Aperture X and Y (mechanical or electrical)	Focus wobbler
4	Focus and stigmation	Focus, stigmator X and Y	Small round features, through focus
5	Checks	None, or a recorded calibration	Tin-on-carbon resolution standard; cross grating

Before you start

- Gun and chamber vacuum are within the normal operating range.
- The specimen is grounded (carbon tape or paint) and, if it is an insulator, coated or imaged below E2 (see SOP-02).
- The stage is at 0° tilt and the specimen at the working distance you will use.
- The beam is on and an image is visible at low magnification.
- You know the type of electron source (table at right).

Terms used

Working distance (WD) is the distance from the pole piece to the focused probe. The **wobbler** swings the objective focus back and forth a few times a second. The **stigmator** is an octupole that cancels two-fold astigmatism with two components, X and Y.

Electron source

Source	Emission	At the start of a session
Schottky FEG	Stable; the tip is always hot (~1800 K)	Check the emission current and the extractor voltage are at their usual values. Nothing to adjust.
Cold FEG	Decreases as gas adsorbs on the tip	Flash the tip when the emission has dropped and at the start of a session.
Tungsten or LaB_6	Set by the filament heating	Saturate the filament; align the gun for maximum current.

Caution. Never vent the gun or raise the extractor voltage of a field-emission gun beyond the service settings. Gun faults are for the facility staff.

Note. Knob and menu names differ between manufacturers and software versions. Where this guide and your facility's procedure or the manufacturer's manual disagree, follow the facility and the manufacturer.

Steps 1 and 2

Gun check and beam conditions

The probe size is a sum of four terms: the demagnified source, spherical and chromatic aberration, and diffraction. Voltage, current, aperture and working distance choose the balance, so set them for the task before aligning.

Step 1 · Gun

1. Read the emission current, extractor voltage and gun vacuum. Compare them with the logbook or the last session.
2. Cold FEG only: if the emission has fallen, flash the tip and wait until the emission is stable.
3. If any value is unusual, stop and call the facility staff.

Step 2 · Beam conditions

1. Choose the voltage. Lower voltage gives more surface detail and less charging; higher voltage gives a smaller probe and more signal.
2. Choose the probe current. Small currents make small probes; analysis (EDS, EBSD, CL) needs nanoamps.
3. Choose the aperture, if it is not set by the current.
4. Set the working distance: short for resolution and the in-lens detector, longer for depth of field and for the analytical detectors.

Typical conditions

Task	Voltage	Current	WD
High-resolution SE (in-lens)	1–5 kV	10–100 pA	2–4 mm
General SE (ETD)	5–15 kV	50–500 pA	5–10 mm
BSE composition	10–20 kV	0.2–2 nA	5–10 mm
Uncoated insulator	below E2 (~1–3 kV)	10–100 pA	3–6 mm
EDS, EBSD, CL	see SOP-02	1–20 nA	analytical WD

Note. Measure the probe current with a Faraday cup before quantitative work: the current display is a calibration, not a measurement.

In SEM-view. The instrument readout shows the probe-size budget (source, C_s, Cc, diffraction) for the current settings; the In-lens mode plots probe size against convergence angle.

Step 3

Aperture centering with the wobbler

If the beam-limiting aperture is off the optic axis, the beam passes through the objective off center. Changing the focus then also tilts the beam, so the image moves sideways as well as blurring, and the probe picks up extra blur.

1. Find a feature with clear edges at medium magnification (about ×5,000–20,000) and focus on it.
2. Turn on the **focus wobbler**. The image pulses in and out of focus.
3. If the image also **sways sideways**, adjust **aperture X** until the sideways movement in that direction stops.
4. Adjust **aperture Y** in the same way.
5. Increase the magnification and repeat with smaller steps.
6. Turn the wobbler off and refocus.

Result. With the wobbler on, the image only pulses in and out of focus about a fixed point.

Note. Some instruments center the aperture electrically (with alignment coils) and some mechanically (with the aperture drive); the wobbler test is the same.

If the image keeps moving. Check it is not stage drift (turn the wobbler off and watch), and that the wobbler amplitude is not so large that the image is lost. Reduce the amplitude at high magnification.

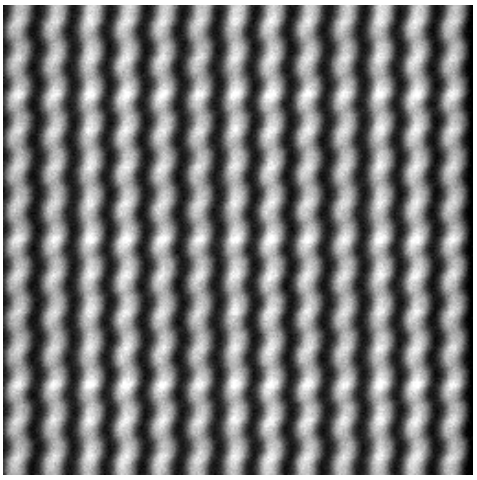


Fig. 1a. Aperture off axis. The two extremes of the wobble, averaged: the image of the cross grating is doubled by the sideways movement.

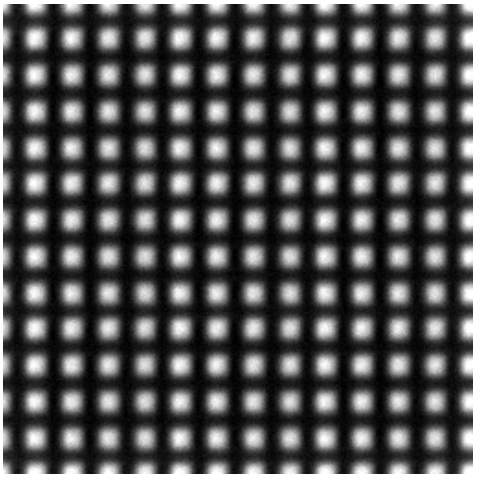


Fig. 1b. Aperture centered. The two extremes overlap; only the focus changes.

Step 4

Focus and stigmation

With astigmatism the lens focuses one direction more strongly than the other, so the probe has two line foci. Just below focus small features streak one way; just above focus they streak at 90° to it. In between they are blurred but not streaked.

1. Choose small, round features (particles, pores) at a magnification where they are a few millimetres across on the screen.
2. Focus as well as you can.
3. Go a little through focus both ways. If the features streak in perpendicular directions either side of focus, there is astigmatism.
4. Return to the best focus and adjust **stigmator X** for the sharpest image.
5. Adjust **stigmator Y** for the sharpest image.
6. Refocus. Repeat steps 3–5 with smaller steps, at a higher magnification if needed, until going through focus no longer streaks the features.

Result. Going through focus blurs the features evenly in all directions, and in focus they are round and sharp.

Note. Stigmate on a slow scan or a reduced area, so the image refreshes quickly but is not too noisy. Auto focus and auto stigmation do the same search with an image-sharpness metric; check their result by eye.

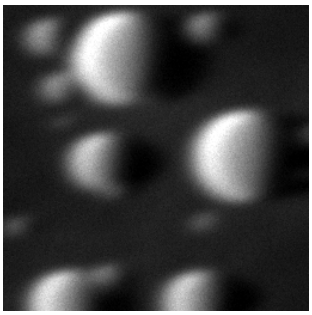


Fig. 2a. Astigmatic, below focus: streaks one way.

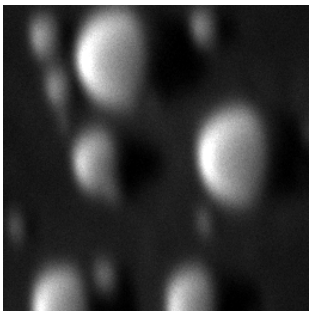


Fig. 2b. Astigmatic, above focus: streaks turned by 90°.

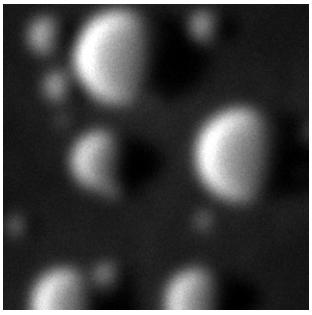


Fig. 2c. Astigmatic, best focus: blurred, not streaked.

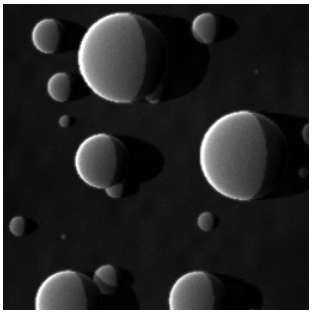


Fig. 2d. Stigmator corrected, in focus.

Step 5

Checks: resolution and magnification

Two standards tell you whether the alignment and the instrument are working as they should: a resolution standard shows the probe size, and a certified grating checks the magnification and the scan.

Resolution

1. Image the tin-on-carbon (or gold-on-carbon) standard at high magnification, with the pixel no larger than a third of the expected probe size.
2. Measure the 25–75 % edge width on several particles, or use the instrument's resolution tool.
3. Compare with the specification for the voltage, current and working distance in use.

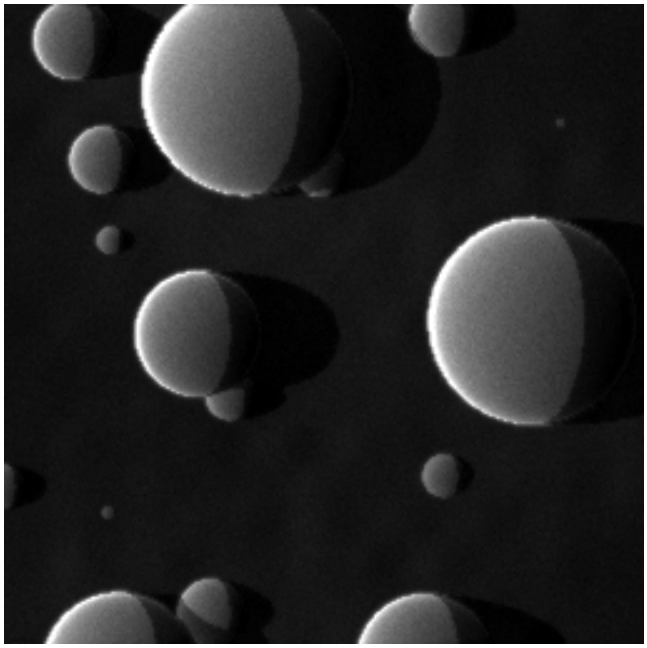


Fig. 3a. Tin on carbon, 400 nm field at 15 kV and 50 pA.

Magnification and scan

1. Image the certified cross grating so that at least four periods fill the field, in both scan directions.
2. Measure the period along x and along y, and the angle between the two families of lines.
3. The ratio of the certified to the measured period is the magnification calibration; the x/y ratio is the scan aspect; the departure from 90° is the scan shear.
4. Record the values and the date; recalibrate after service or when the measured error exceeds the facility's tolerance (often 1–2 %).

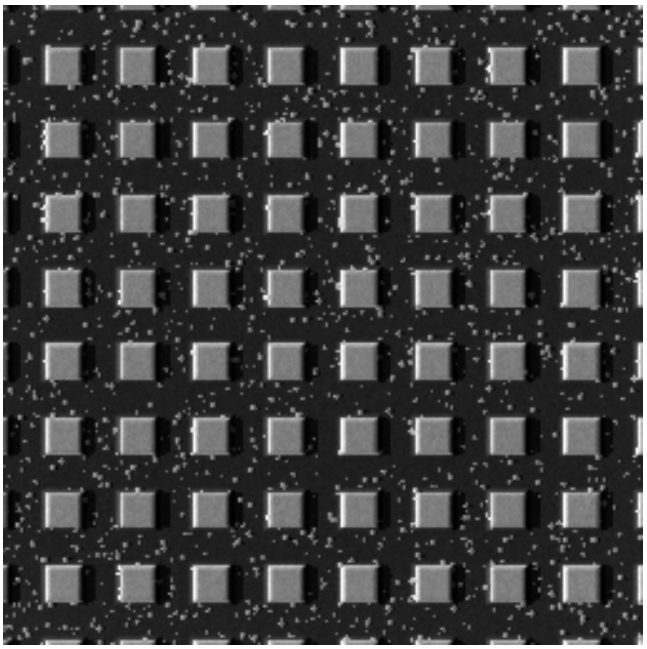


Fig. 3b. 463 nm cross grating (2160 lines/mm), 4 µm field.

In SEM-view. Say or type “measure the resolution” or “calibrate the magnification”, or use Calibrate from data in the Digital twin section: the fits recover the resolution, magnification, scan aspect and shear and compare them with the simulator's truth.

Practice on SEM-view

The figures in this procedure were computed with SEM-view, a physics-based SEM simulator that runs in a web browser. Its training tasks “Focus and stigmate” and “Center the aperture” hide a random misalignment and report how close you came once you are done.

Open the training task

advanced-em.github.io/SEM-view/?train=sem-focus-stig

Scan the code with a phone, or open SEM-view and select Training.

Simulated column

Select any part of the 3D instrument, from the Schottky gun to the detectors, to see what it does and how it is measured.

Scored practice

Rate your own alignment, then compare it with the hidden misalignment.

Research data

With your consent, anonymous practice sessions are used to study how people align microscopes.

SOP-01 SEM column alignment, version 1.0, October 2026
For training. Follow your facility's procedures and the manufacturer's documentation.
github.com/Advanced-EM/SEM-view

Scanning with Electrons

SEM-view · Advanced-EM